

TC2343

LAB03 ROS TOPIC

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FTSM UKM

Reference

eBOOK ROS Robot Programming – Chapter 7 Basic ROS Programming (pg 174)

Instructions

1. (C++) Create and Run Publisher and Subscriber Nodes (Follow instructions from page 151 until 162)
 - a. Creating a Package
 - b. Modifying the Package Configuration File (package.xml)
 - c. Modifying the Build Configuration File (CMakeLists.txt)
 - d. Writing the Message File
 - e. Writing the Publisher Node
 - f. Writing the Subscriber Node
 - g. Building a Node
 - h. Running the Publisher
 - i. Running the Subscriber
 - j. Checking the Communication Status of the Running Nodes
2. Create and Edit ROSLaunch (Follow instruction from page 189 until 193)
 - a. Using the roslaunch to run both publisher and subscriber
 - b. Launch tag
3. (C++) Write a publisher node to move the Turtlesim and then Turtlebot3.
 - a. Make the turtlesim to draw square
https://docs.ros.org/en/api/turtlesim/html/draw_square_8cpp_source.html
 - b. Make the Turtlebot3 waffle to draw square in turtlebot3_world (using previous lab simulation)

Snapshot all the task and submit as Assignment 3 in UKMFolio

“HAPPY ROBOTING”

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